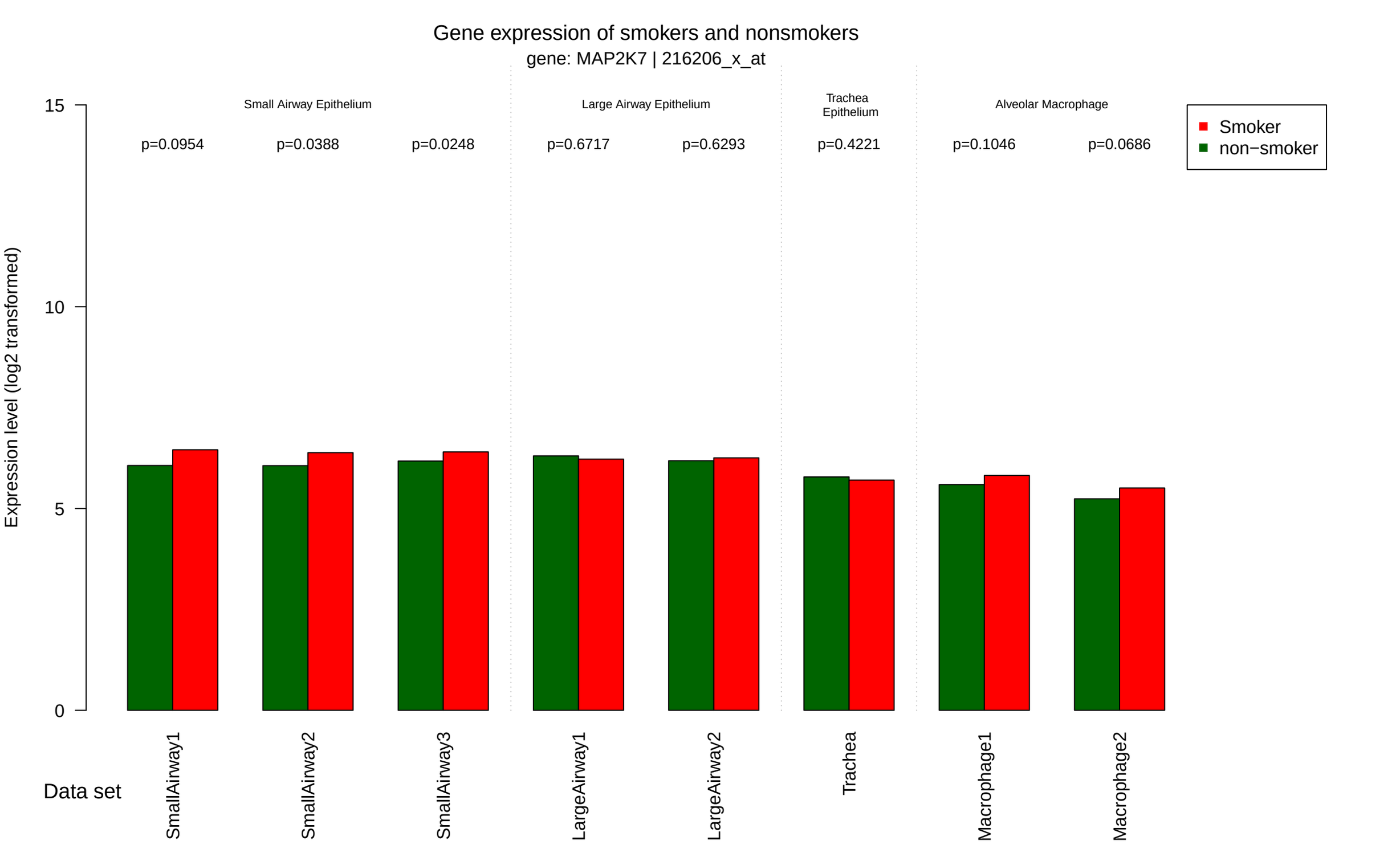
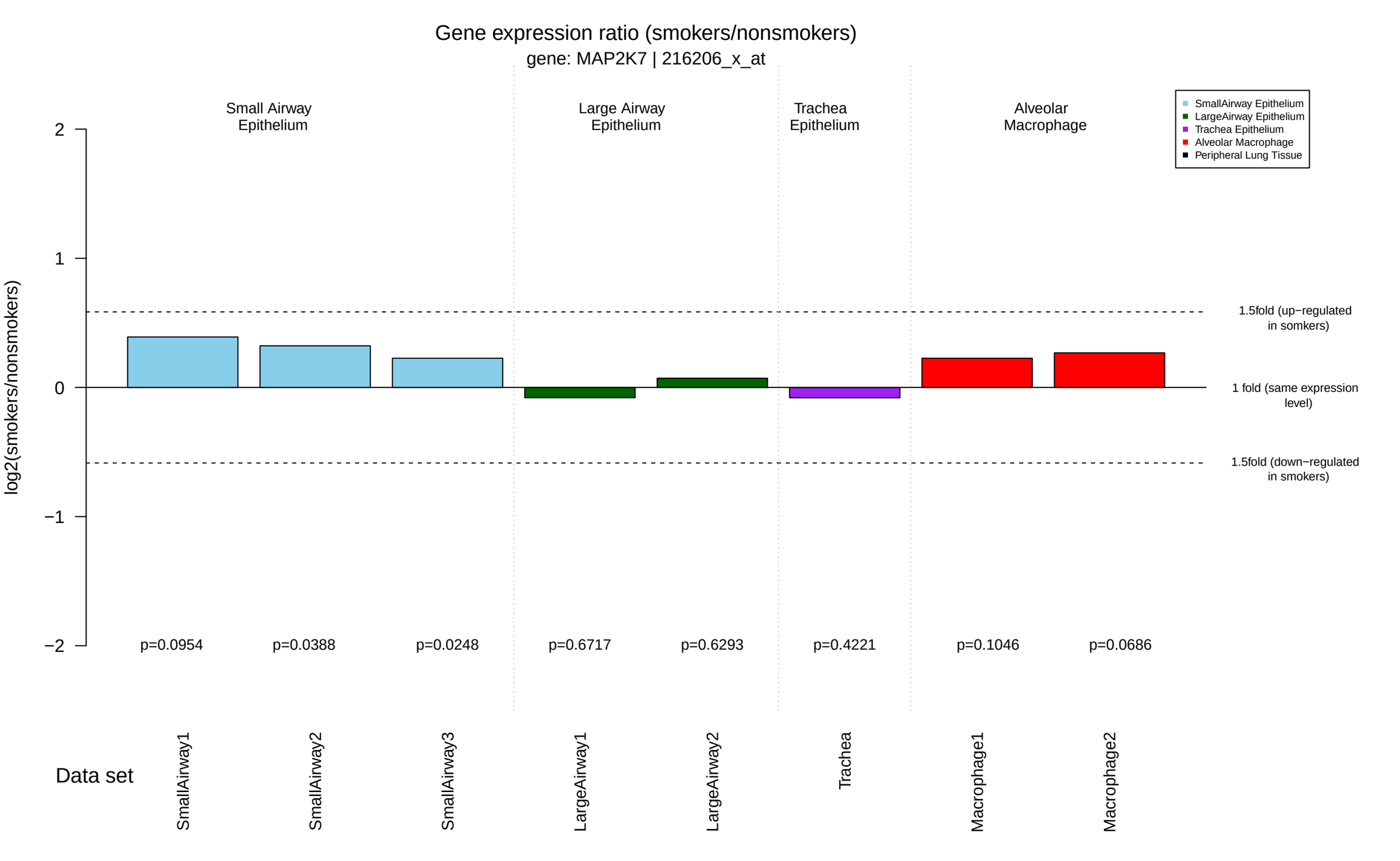
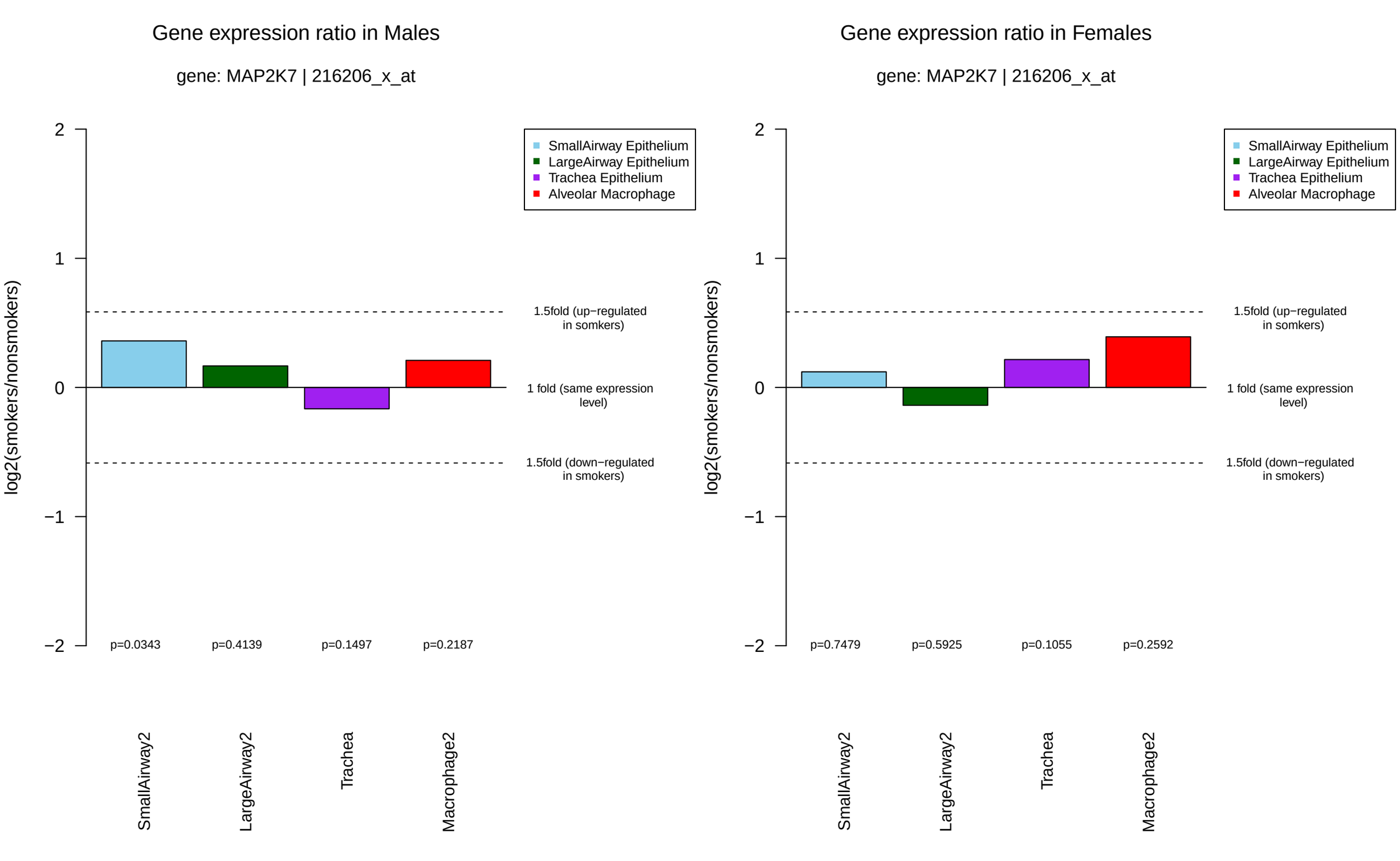
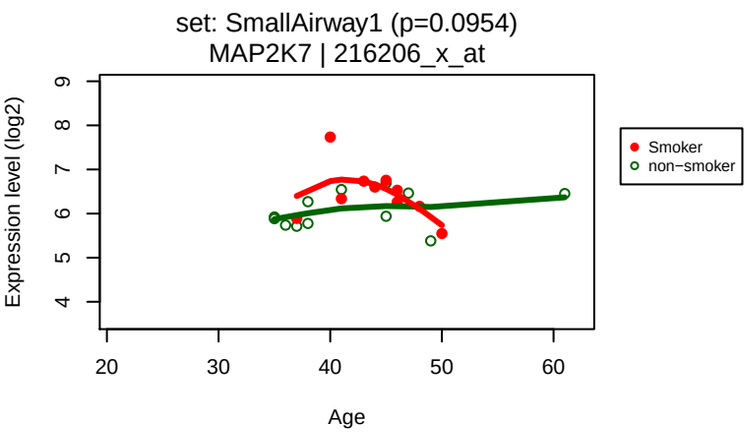
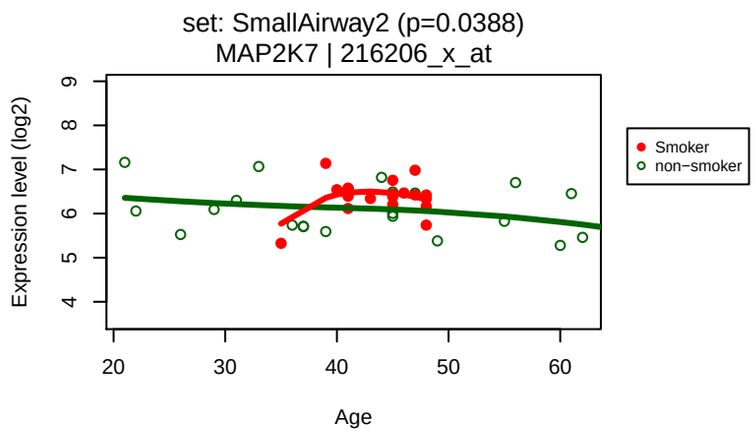
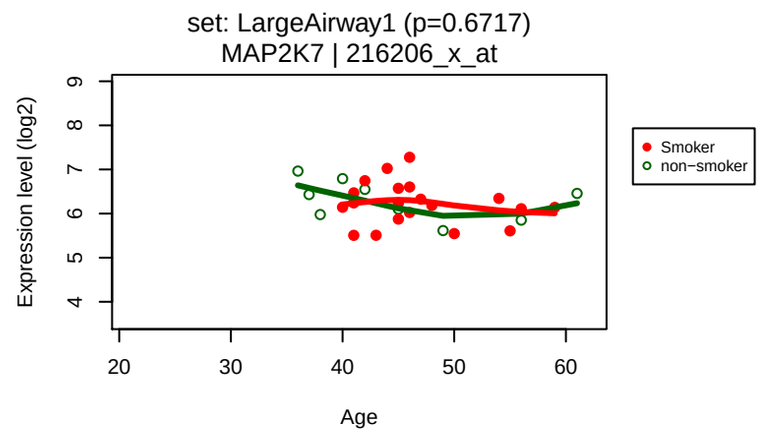

Gene expression of smokers and nonsmokers
gene: MAP2K7 | 216206_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0954
p=0.0388
p=0.0248
p=0.6717
p=0.6293
p=0.4221
p=0.1046
p=0.0686
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

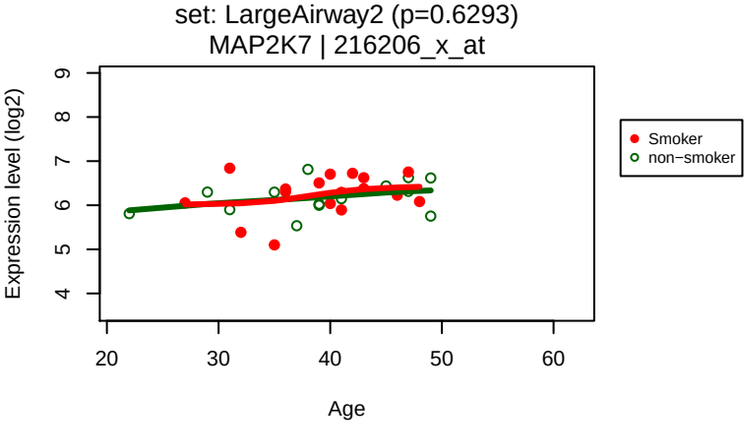

Gene expression ratio (smokers/nonsmokers)
gene: MAP2K7 | 216206_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0954
p=0.0388
p=0.0248
p=0.6717
p=0.6293
p=0.4221
p=0.1046
p=0.0686
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

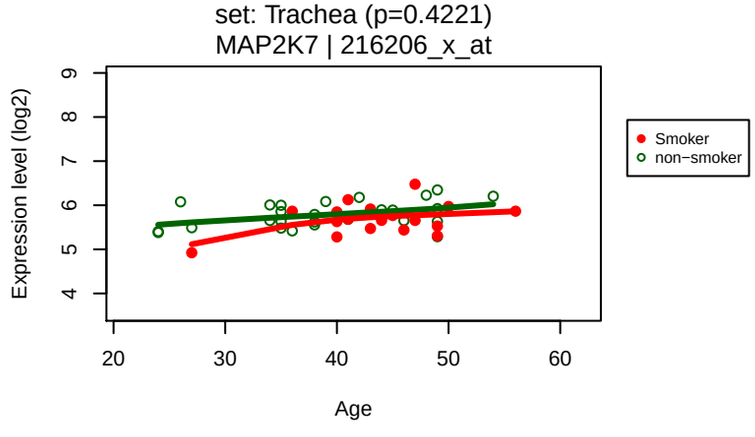

Gene expression ratio in Males
gene: MAP2K7 | 216206_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0343
p=0.4139
p=0.1497
p=0.2187
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAP2K7 | 216206_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7479
p=0.5925
p=0.1055
p=0.2592
SmallAirway2
LargeAirway2
Trachea
Macrophage2

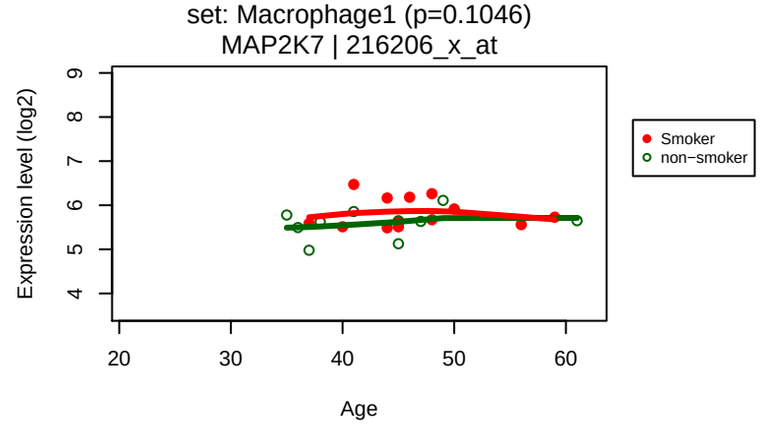

set: SmallAirway1 (p=0.0954)
MAP2K7 | 216206_x_at
Expression level (log2)
Age
Smoker
non−smoker

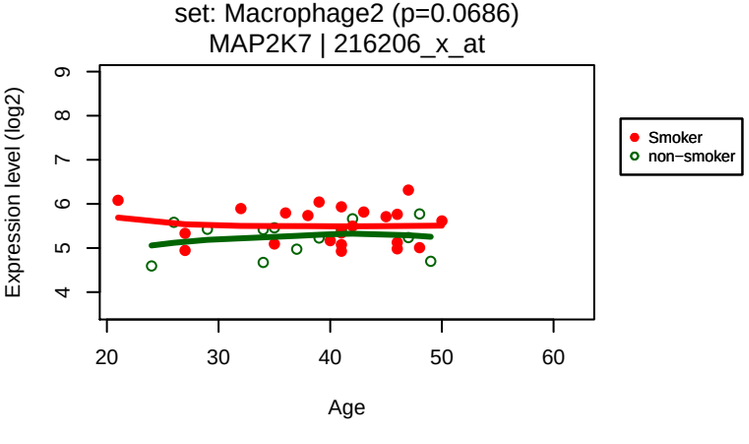

set: SmallAirway2 (p=0.0388)
MAP2K7 | 216206_x_at
Expression level (log2)
Age
Smoker
non−smoker

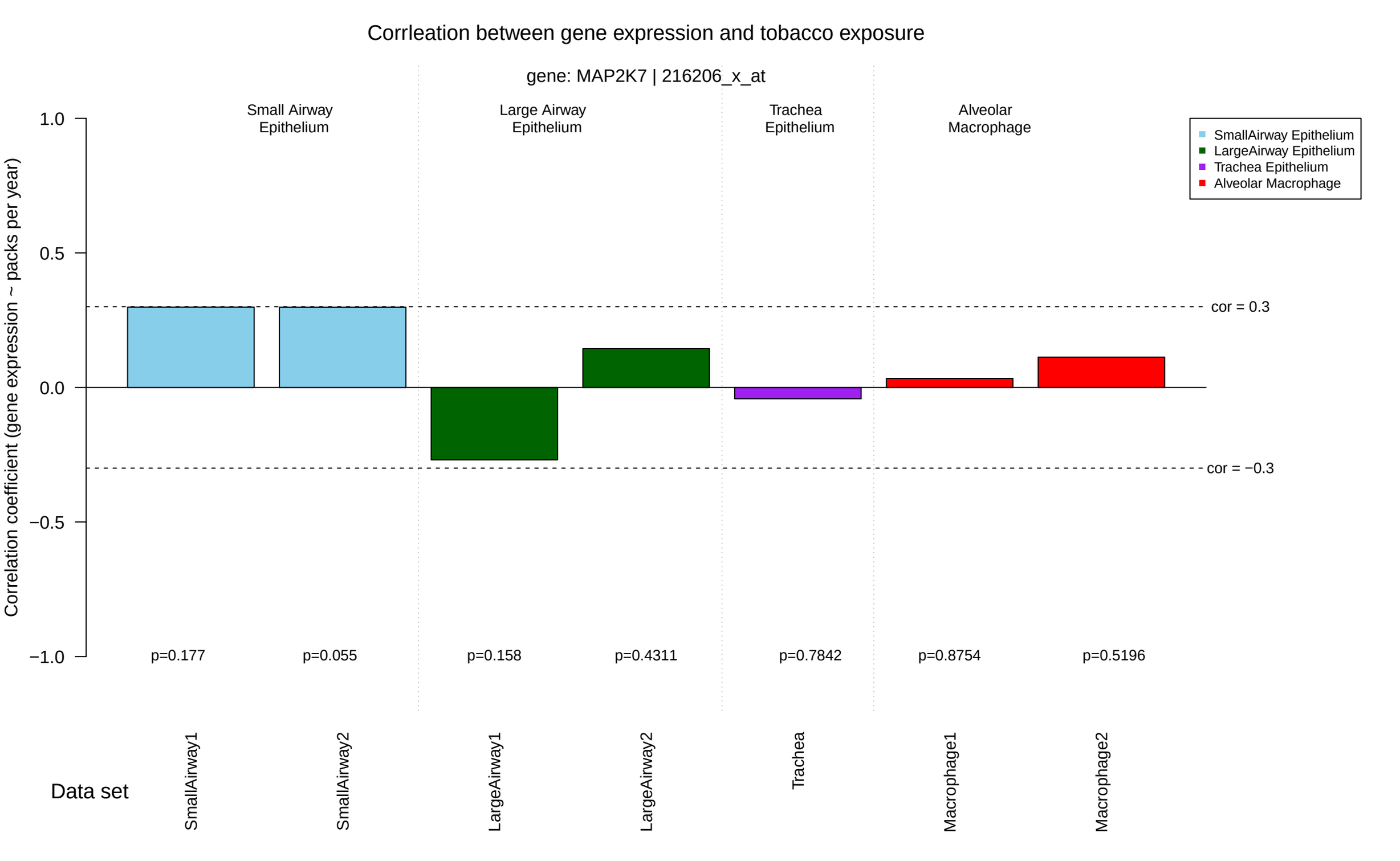

set: LargeAirway1 (p=0.6717)
MAP2K7 | 216206_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6293)
MAP2K7 | 216206_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4221)
MAP2K7 | 216206_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1046)
MAP2K7 | 216206_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0686)
MAP2K7 | 216206_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAP2K7 | 216206_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.177
p=0.055
p=0.158
p=0.4311
p=0.7842
p=0.8754
p=0.5196
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2